



# Categorical Data Analysis

Presented by

王媛鈺

吳思蓓

徐淳姿



# OUTLINE

## Introduction

- Categorical Data
- Probability Distribution
- Test statistics
- Contingency Table
- ROC/AUC
- Comparing Proportions
- Independence test

## GLM

- Comparison of LM and GLM
- GLM R code outcome
- Model Testing

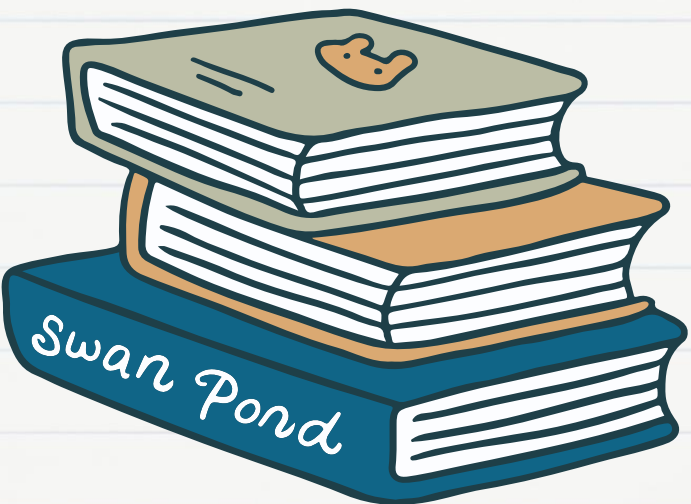
## GEE

- McNemar test
- Generalized Estimating Equations



# Introduction

R26124021 王媛鈺



# Categorical Data (Plot in R.)

Two types of  
categorical  
variable

**Nominal:**  
unordered categorical

**Ordinal:**  
ordered categorical  
e.g. frequency,  
satisfaction

The response variable  
⇒ **Y** variable  
The explanatory variable  
⇒ **X** variable

**Y variable**  
(Yes, No)  
(excellent, good, fair, poor)  
(Apple, Samsung, LG, others)

**X variable**  
**categorical or continuous**  
Income, age, gender, education, race, ...

# Binomial Distribution



- n independent Bernoulli trials

- two possible outcomes for each trial  $\Rightarrow$  (success, failure).

- $p = P(\text{success}), 1 - p = P(\text{failure})$

- $Y =$  numbers of success in n trials

- $Y \sim \text{Bin}(n, p)$

$$\Rightarrow P(y) = \frac{n!}{y!(n-y)!} p^y (1-p)^{n-y}, \quad y = 0, 1, 2, 3, \dots$$



# Multinomial Distribution



- ▣ When each trail has more than 2 outcomes.
- ▣ Let  $c$  be the number of outcome in categories
- ▣ Probability  $\Rightarrow (p_1, p_2, p_3, \dots, p_c), \sum_{i=1}^c p_i = 1$
- ▣ For  $n$  independent observations,  $\sum_{j=1}^c y_j = n$
- ▣  $\Rightarrow P(y_1, y_2, \dots, y_c) = \frac{n!}{y_1! y_2! \dots y_c!} p_1^{y_1} p_2^{y_2} \dots p_c^{y_c}, \quad \sum_{j=1}^c y_j = n$





M<sub>aximum</sub> L<sub>ikelihood</sub>

# Inference about Binomial

Estimates:  $\hat{p} = p = \frac{y}{n}$

Expectation:  $E(\hat{p}) = p$ ;

Variance:  $\sigma(\hat{p}) = \sqrt{\frac{p(1-p)}{n}}$

Test statistic

$$H_o: p = p_o$$

$$H_a: p \neq p_o$$

$$Z = \frac{\hat{p} - p_o}{\sqrt{\frac{p_o(1-p_o)}{n}}} \sim N(0,1)$$

$$\text{P-value} = 2 \times P(|Z| > Z_{\alpha/2})$$

Example:

Do a majority, or minority, of adults in the United States believe that a pregnant woman should be able to obtain an abortion?

Let  $p$  denote the proportion of the American adult population that responds “yes” to the question. Responses to survey of 1810 answers, 837 replied “yes” and 973 replied “no”.

We test:

$$\begin{cases} H_o: p = 0.5 \\ H_a: p \neq 0.5 \end{cases}, \hat{p} = \frac{837}{1810} = 0.462$$

$$Z = \frac{\hat{p} - p_o}{\sqrt{\frac{p_o(1-p_o)}{n}}} = \frac{0.462 - 0.5}{\sqrt{\frac{0.5(1-0.5)}{1810}}} = -3.2 ; \text{P-value} = 0.0014 < 0.05$$

We reject null hypothesis at significant level of 0.05, that is, fewer than half of Americans favored unrestricted legal abortion.

# Confidence Interval for Binomial Parameter

▫ **Wald CI** for a parameter  $\theta$  is  $\hat{\theta} \pm Z_{\alpha/2}(SE) \Rightarrow \hat{p} \pm Z_{.025} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$

$$\Rightarrow \text{e.g. } \hat{p} \pm Z_{.025} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} = 0.462 \pm 1.96 \sqrt{\frac{0.462 \times 0.538}{1810}} = (0.439, 0.485).$$

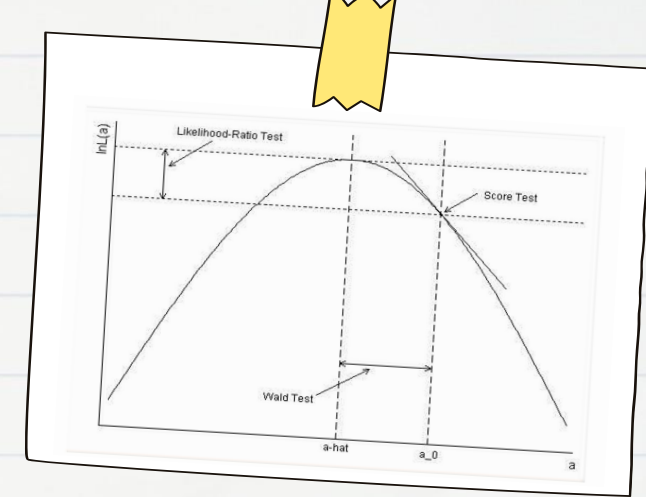
Note:

▫ When  $p$  is near 0 or near 1, it performs poorly unless  $n$  is large enough.



# Wald/Score/LRT

- All three tests use the **likelihood** of the models being compared to assess their fit.



|                                | When/How to use                                                                                             | Perspective 檢定的觀點                                                                 |
|--------------------------------|-------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| Wald                           | By testing $H_0$ that a set of parameters equal to zero or not.                                             | Distance between <b>a_0</b> and <b>a-hat</b> on the x-axis                        |
| Score                          | By testing the expected change in slope of the likelihood function.                                         | Looks at the slope of the log likelihood when <b>a</b> is constrained to be zero. |
| LRT<br>Likelihood<br>RatioTest | By testing the log-likelihood of <u>2 models</u> with values of the parameter <b>a</b> constrained to zero. | Comparing the height of the likelihoods for the two models                        |

- **Extend to model:**
  - **Testing Goal: Does constraining parameters to 0 reduces the fit of model?**  
 $\Rightarrow H_0: \theta = 0$
  - As the sample size becomes large, the result will be approximately the same.

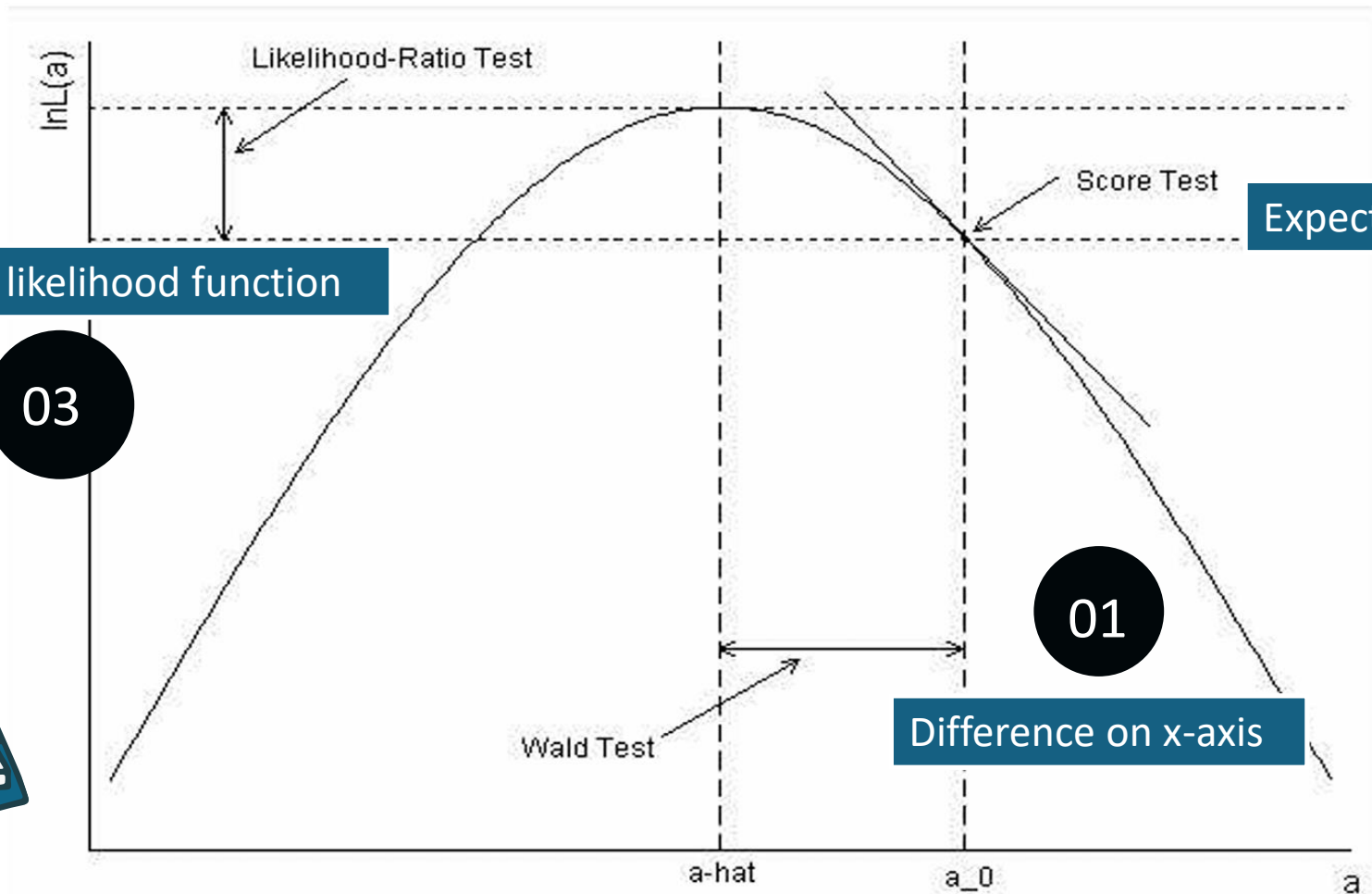


# Wald/Score/LRT



Heights of likelihood function

03



02

Expected change on slope

01

Difference on x-axis

# Contingency table

- A table that cross classifies two variables is call two-way contingency table
- Take diagnostic disease test for example:



|           |   | Y(predict) |    |
|-----------|---|------------|----|
|           |   | 1          | 2  |
| X(Actual) | 1 | TP         | FN |
|           | 2 | FP         | TN |

- **True positive (TP):** Prediction is + and X is +.

- **True negative (TN):** Prediction is – and X is –.

- **False positive (FP):** Prediction is + and X is –.

- **False negative (FN):** Prediction is– and X is +.

$$\square \text{Accuracy} = \frac{TP+TN}{TP+FP+FN+TN}$$

$$\square \text{Precision} = \frac{TP}{TP+FP}$$

▫ F1 score

$$= \frac{2(\text{sensitivity} \times \text{Precision})}{(\text{sensitivity} + \text{Precision})}$$

⇒ **Sensitivity** =  $P(Y=1 | X=1)$

The predicted outcome is positive, given the actual situation is positive.

⇒ **Specificity** =  $P(Y=2 | X=2)$

The predicted outcome is negative, given the actual situation is negative.

# ROC curve construction

## □ Scatter plot

e.g.  $Y: \text{obese}(0,1) \sim x(\text{weight})$

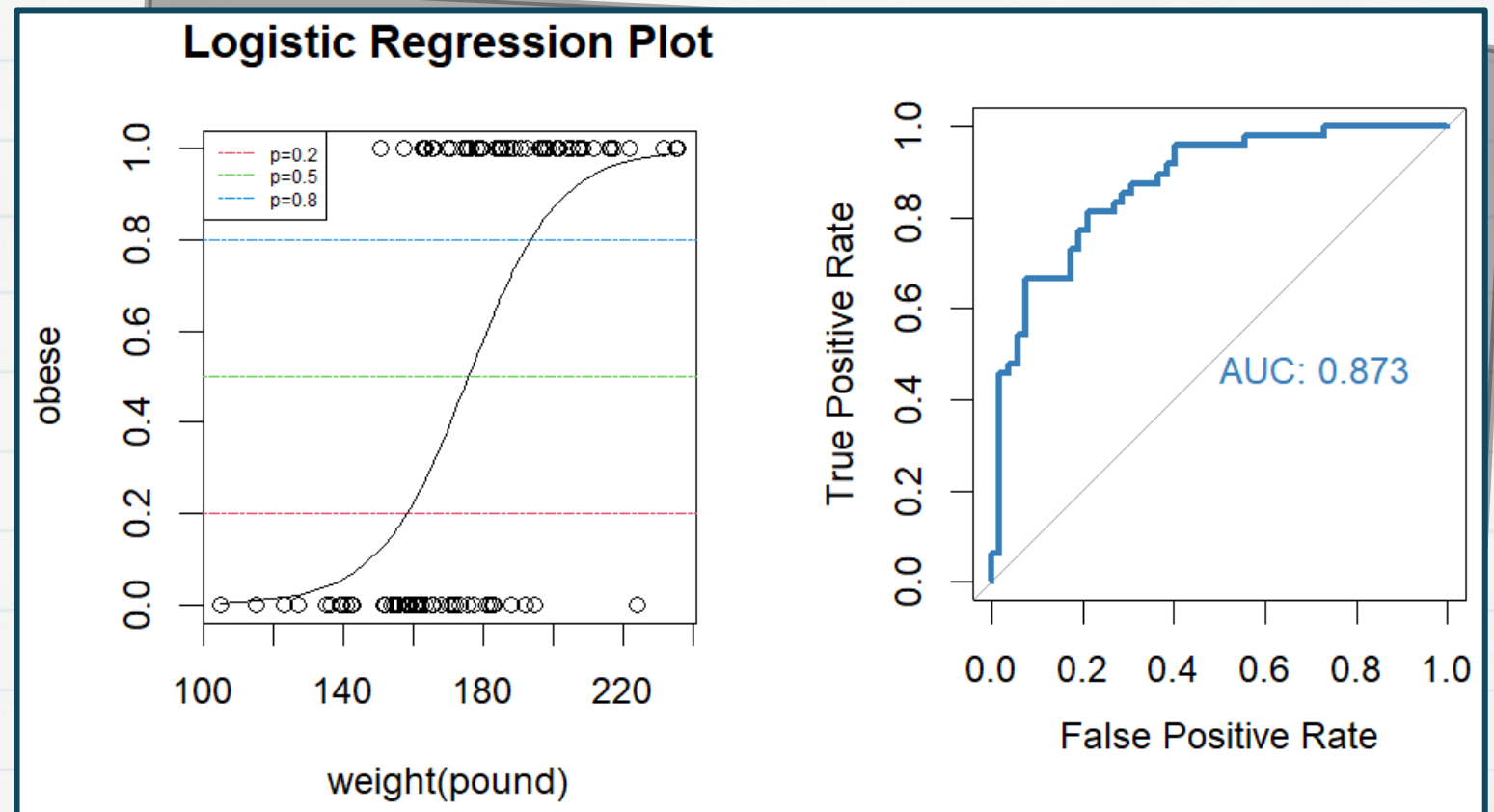
## □ ROC curve

- Receiver operating characteristic curve
- Summarization of tons of confusion matrices

## □ Cut point(Threshold)

## □ AUC

- Area Under Curve
- $\text{AUC} = 0.5$
- $0.6 < \text{AUC} < 0.8$
- $\text{AUC} > 0.8$



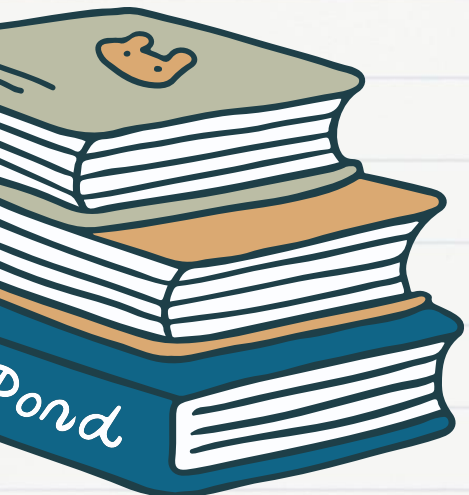
# Comparing proportion in 2-2 table

- Difference of proportion
- Ratio of proportions, Relative risk
- Odds ratio

|  | Y      |         |             |
|--|--------|---------|-------------|
|  |        | Success | Failure     |
|  | X      |         |             |
|  | Group1 | $p_1$   | $(1 - p_1)$ |
|  | Group2 | $p_2$   | $(1 - p_2)$ |

$\hat{p}_1$  = the sample proportions of successes for group 1.

$\hat{p}_2$  = the sample proportions of successes for group 2.



# Difference of proportion

|   |         | Y(Heart Attack) |       |                |                      |
|---|---------|-----------------|-------|----------------|----------------------|
| X |         | Yes             | No    | P(Yes)         | P(No)                |
|   | Placebo | 189             | 10845 | $p_1 = 0.0171$ | $(1 - p_1) = 0.9829$ |
|   | Aspirin | 104             | 11037 | $p_2 = 0.0094$ | $(1 - p_2) = 0.9906$ |

$\hat{p}_1$  = the sample proportions of successes for group 1.

$\hat{p}_2$  = the sample proportions of successes for group 2.

**Example: Physicians Health Study**

$$H_o: p_1 = p_2$$

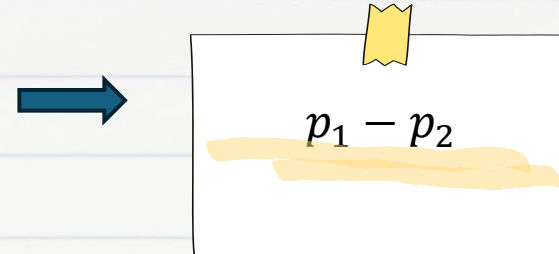
$$H_a: p_1 \neq p_2$$

$$\hat{p}_1 - \hat{p}_2 = p_1 - p_2 = 0.017 - 0.009 = 0.008$$

$$SE = \sqrt{\frac{0.017 \times 0.983}{11034} + \frac{0.009 \times 0.991}{11037}} = 0.0015$$

$$95\% \text{ CI for } p_1 - p_2 = (0.005, 0.011) \Rightarrow p_1 - p_2 > 0 \Rightarrow p_1 > p_2$$

Conclusion : Taking aspirin appears to result in a diminished risk of heart attack.



# Ratio of proportion/ Relative Risk

|   |         | Y(Heart Attack) |       |                |                      |
|---|---------|-----------------|-------|----------------|----------------------|
| X |         | Yes             | No    | P(Yes)         | P(No)                |
|   | Placebo | 189             | 10845 | $p_1 = 0.0171$ | $(1 - p_1) = 0.9829$ |
|   | Aspirin | 104             | 11037 | $p_2 = 0.0094$ | $(1 - p_2) = 0.9906$ |

$\hat{p}_1$  = the sample proportions of successes for group 1.

$\hat{p}_2$  = the sample proportions of successes for group 2.



$$\frac{p_1}{p_2}$$

## Example: Physicians Health Study

$$H_0: p_1 = p_2$$

$$H_a: p_1 \neq p_2$$

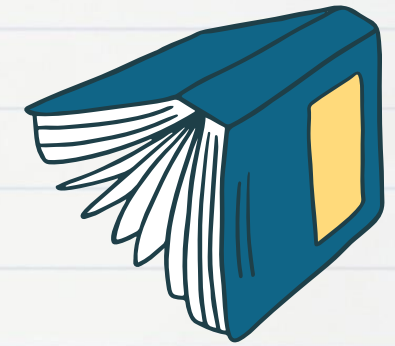
$$\hat{p}_1 = \frac{p_1}{p_2} = \frac{0.017}{0.009} = 1.89$$

$$95\%CI \Rightarrow (1.43, 2.31)$$

⇒ Sample proportion of heart attacks was 89% higher for placebo group.

Note:  $H_0: p_1 = p_2$ , if groups are independent  $\leftrightarrow \frac{p_1}{p_2} = 1$





# Odds ratio

|   |         | Y(Heart Attack) |       |                |                      |
|---|---------|-----------------|-------|----------------|----------------------|
| X |         | Yes             | No    | P(Yes)         | P(No)                |
|   | Placebo | 189             | 10845 | $p_1 = 0.0171$ | $(1 - p_1) = 0.9829$ |
|   | Aspirin | 104             | 11037 | $p_2 = 0.0094$ | $(1 - p_2) = 0.9906$ |

$\hat{p}_1$  = the sample proportions of successes for group 1

$\hat{p}_2$  = the sample proportions of successes for group 2




$$\text{Odds Ratio: } \theta = \frac{p_1 / (1 - p_1)}{p_2 / (1 - p_2)}$$

$$\text{Odds}_1 = \frac{0.0171}{0.9829} = 0.0174$$

$$\text{Odds}_2 = \frac{0.0094}{0.9906} = 0.0095$$


$$\hat{\theta} = \frac{0.0174}{0.0095} = 1.83$$

⇒ The odds of a heart attack for placebo group were 1.83 time odds for aspirin group.



# Inference of odds ratio

|   |         | Y(Heart Attack) |       |                |                      |
|---|---------|-----------------|-------|----------------|----------------------|
| X |         | Yes             | No    | P(Yes)         | P(No)                |
|   | Placebo | 189             | 10845 | $p_1 = 0.0171$ | $(1 - p_1) = 0.9829$ |
|   | Aspirin | 104             | 11037 | $p_2 = 0.0094$ | $(1 - p_2) = 0.9906$ |

$$\begin{aligned} \square \text{ SE}(\log \hat{\theta}) &= \sqrt{\frac{1}{n_{11}} + \frac{1}{n_{12}} + \frac{1}{n_{21}} + \frac{1}{n_{22}}} \\ \square \text{ CI for } \log \theta &= \log \hat{\theta} \pm Z_{\alpha/2} \text{SE}(\log \hat{\theta}) \\ \square \text{ CI for } \theta &= (e^L, e^U) \end{aligned}$$

## Example: Physicians Health Study

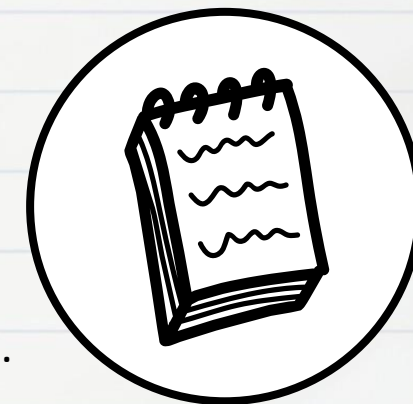
$$95\% \text{ CI for } \log \theta = 0.605 \pm 1.96 \sqrt{\frac{1}{189} + \frac{1}{10933} + \frac{1}{104} + \frac{1}{10845}} = (0.356, 0.846)$$

$$95\% \text{ CI for } \theta = (e^{0.356}, e^{0.846}) = (1.44, 2.33) > \theta = 1$$

Which does not cover  $\theta = 1$ , that is, the Y and X are significantly correlated.



# Note



- **Proportion difference vs. R.R.**

- Using the difference between proportions might be misleading when the value is close to 0.
- E.g. difference between (0.410, 0.401) and (0.01, 0.001)

- **R.R. vs. Odds Ratio**

- **R.R.**  $\Rightarrow$  Fixed group size, 隨機試驗 randomized trial, 世代研究 cohort study  $\Rightarrow$  **Experimental study**
- **O.R.**  $\Rightarrow$  前瞻性研究 Prospective study, 回溯性研究 retrospective study, and 對照型研究 Case-control study  $\Rightarrow$  **Observational study**

- While computing **odds ratio**, we should be aware of “sparse”

- Hence we add 0.5 at cell whose value is 0.
- E.g.

|   | Y  |    |
|---|----|----|
| X | 10 | 0  |
|   | 23 | 18 |

 $\rightarrow$ 

|   | Y  |     |
|---|----|-----|
| X | 10 | 0.5 |
|   | 23 | 18  |



# Test of Independence: Chi-squared Test

$H_0$ : X and Y are independent

$H_a$ : X and Y are not independent

| Gender  | Political Party Identification |            |             | Total |
|---------|--------------------------------|------------|-------------|-------|
|         | Democrat                       | Republican | Independent |       |
| Females | 495(456.9)                     | 272(297.4) | 590(602.6)  | 1357  |
| Males   | 330(368.1)                     | 265(239.6) | 498(485.4)  | 1093  |
| Total   | 825                            | 537        | 1088        | 2450  |

$$\chi^2 = \sum_{all\ cells} \frac{(n_{ij} - \hat{\mu}_{ij})^2}{\hat{\mu}_{ij}} \sim \chi^2_{df=(i-1)(j-1)} \quad \text{where } \hat{\mu}_{ij} = \frac{n_{i+}n_{+j}}{n}$$

We reject  $H_0$  if p-value < 0.05

$$\Rightarrow \chi^2 = 12.57, \text{ p-value}=0.002$$



# Test of Independence: for ordinal data

- For X's scores:  $x_1 \leq x_2 \leq \dots \leq x_i$

- For Y scores:  $y_1 \leq y_2 \leq \dots \leq y_j$

$H_o$ : X and Y are independent

$H_a$ : X and Y are not independent

- Calculate sample correlation R between X and Y

$$-1 \leq R = \frac{\sum_{i,j} (x_i - \bar{x})(y_j - \bar{y}) \hat{p}_{ij}}{[\sum_i (x_i - \bar{x})^2 p_{i+}][\sum_j (y_j - \bar{y})^2 p_{+j}]} \leq 1$$

Test statistics:  $M^2 = (n - 1)R^2 \sim \chi^2_{df=1}$

We reject  $H_o$  if p-value < 0.05.



# Test of Independence: Fisher Exact Test

- $2 \times 2$  case (Fisher 1935)

|          |          |          |
|----------|----------|----------|
| $n_{11}$ | $n_{12}$ | $n_{1+}$ |
| $n_{21}$ | $n_{22}$ | $n_{2+}$ |
| $n_{+1}$ | $n_{+2}$ | $n$      |

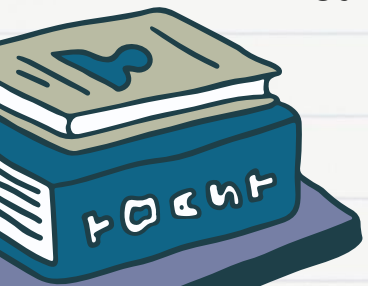
Note:

- For row and column sum are all fixed.
- No need distribution assumptions.
- Sample size is small or unbalanced.

Exact null distribution of  $\{n_{ij}\}$  is based on **fixed row** and **column** totals, which is

$$P(n_{11}) = \frac{\binom{n_{1+}}{n_{11}} \binom{n_{2+}}{n_{+1}-n_{11}}}{\binom{n}{n_{1+}}} \text{ (Hypergeometric distribution)}$$

- Calculate the **extreme probability value** of cell  $n_{11}$  by hypergeometric distribution.
- If the extreme probability is less than significant level, and then we reject null hypotheses.



# Fisher Exact Test: Example

|                  |      | Guess(Y) |     |   |
|------------------|------|----------|-----|---|
|                  |      | Milk     | Tea |   |
| Poured First (X) | Milk | 3        | 1   | 4 |
|                  | Tea  | 1        | 3   | 4 |
|                  |      | 4        | 4   | 8 |

$H_0$ : X and Y are independent  $\theta = 1$

$H_a$ : X and Y are not independent  $\theta \neq 1$

$$P(n_{11} \geq 3) = \frac{\binom{4}{3}\binom{4}{1} + \binom{4}{4}\binom{4}{0}}{\binom{8}{8}} = 0.014 + 0.229 = 0.243$$

Since  $0.243 > 0.05$

We do not reject  $H_0$ , that is, the Milk Tea Lady might not be able to distinguish the difference of which drink is poured first.

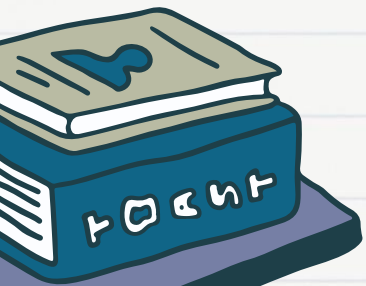
|   |   |
|---|---|
| 4 | 0 |
| 0 | 4 |

|   |   |
|---|---|
| 3 | 1 |
| 1 | 3 |

|   |   |
|---|---|
| 2 | 2 |
| 2 | 2 |

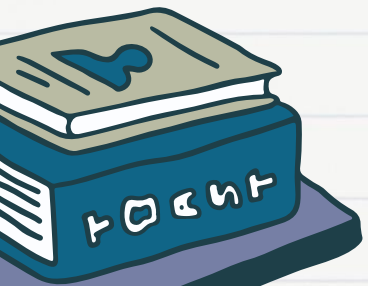
|   |   |
|---|---|
| 1 | 3 |
| 3 | 1 |

|   |   |
|---|---|
| 4 | 0 |
| 0 | 4 |



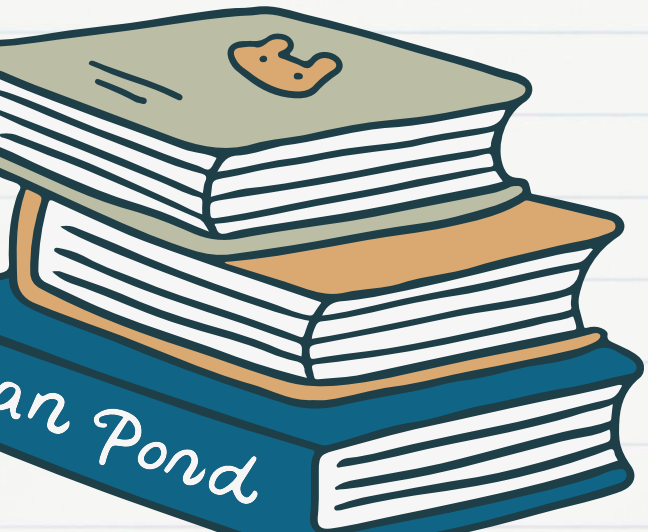
# Reference

- <https://stats.oarc.ucla.edu/other/mult-pkg/faq/general/faqhow-are-the-likelihood-ratio-wald-and-lagrange-multiplier-score-tests-different-and-or-similar/>
- <https://www.iwh.on.ca/what-researchers-mean-by/observational-vs-experimental-studies>
- <https://www.youtube.com/watch?v=4jRBRDbJemM>
- <https://www.youtube.com/watch?v=qcvAqAH60Yw>



# Generalized Linear Model (GLM)

R26121065 吳思蓓





# Introduction

- Comparison of LM and GLM
- GLM R code outcome
- Model Testing

# Why **use** GLM?



- For these questions :
    - 1) If I have data such as season, time, surface temperature, etc., how can I estimate the concentration of PM2.5?
    - 2) If I have basic information about a car, how can I estimate its selling price?
    - 3) If I have basic information about students, how can I estimate their exam scores?
- Use LM(Linear Model)

# But For Categorical Data

- We have these question :
  - 1) If I have basic information about patients, how can I estimate the **probability** of a patient developing diabetes?
  - 2) If I have basic information about students, how can I determine **which course category** they will choose in the future?
  - 3) If I have basic information about students, how can I predict **how many awards** a student will receive in a year?

→ Use GLM



# Comparison of LM(Linear Model) and GLM



|                         | LM                                                                                           | GLM                                                                                                               |
|-------------------------|----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Purpose                 | Predict Y                                                                                    |                                                                                                                   |
| Explanatory Variable(X) | Continuous/Discrete                                                                          |                                                                                                                   |
| Response Variable(Y)    | Continuous/Discrete                                                                          |                                                                                                                   |
| Estimation Method       | Use LM<br>$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \epsilon_{ij}$<br>to estimate Y | $g(\mu) = \beta_0 + \beta_1 x_1 + \dots$<br>$Y \sim f_Y(y \mu), E(Y) = \mu$<br>$g(\mu) \rightarrow$ link function |
| Range of Y              | $-\infty < y < \infty$<br>(Extrapolation not allowed)                                        | By the model we select                                                                                            |
| Residual Assumption     | $\epsilon_{ij} \sim iid N(0, \sigma^2)$                                                      | None                                                                                                              |
| Example                 | Mileage of the car                                                                           | Whether involved in an accident,<br>number of accidents                                                           |

- R code outcome

Call: Response Variable Explanatory Variable Model and link function  
`glm(formula = num_awards ~ math + C.prog, family = poisson(link = "log"), data = awards)`

Deviance Residuals:

| Min     | 1Q      | Median  | 3Q     | Max    |
|---------|---------|---------|--------|--------|
| -2.2043 | -0.8436 | -0.5106 | 0.2558 | 2.6796 |

Checking for abnormal residuals

Coefficients:

|             | Estimate | Std. Error | z value | Pr(> z )     |
|-------------|----------|------------|---------|--------------|
| (Intercept) | -5.24712 | 0.65845    | -7.969  | 1.60e-15 *** |
| math        | 0.07015  | 0.01060    | 6.619   | 3.63e-11 *** |
| C.progB     | 1.08386  | 0.35825    | 3.025   | 0.00248 **   |
| C.progC     | 0.36981  | 0.44107    | 0.838   | 0.40179      |

Significance of coefficients

---  
 Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

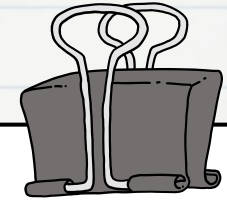
(Dispersion parameter for poisson family taken to be 1)

|                           |        |                    |
|---------------------------|--------|--------------------|
| Null deviance: 287.67     | on 199 | degrees of freedom |
| Residual deviance: 189.45 | on 196 | degrees of freedom |
| AIC: 373.5                |        |                    |

Used to test the significance of the model.

Number of Fisher Scoring iterations: 6

# Model Testing



- No distributional assumption  
→ Assume  $\beta_i$  follows a normal distribution using the properties of large sample normality.

- Testing the significance of coefficients → Wald-test

$$\begin{cases} H_0 : \beta_i = 0 \\ H_a : \beta_i \neq 0 \end{cases}$$

$$z = \frac{\widehat{\beta}_i}{SE_{\beta_i}} \sim N(0,1)$$

- Testing the significance of the model → Likelihood Ratio Test

$$-2(L_0 - L_1)$$

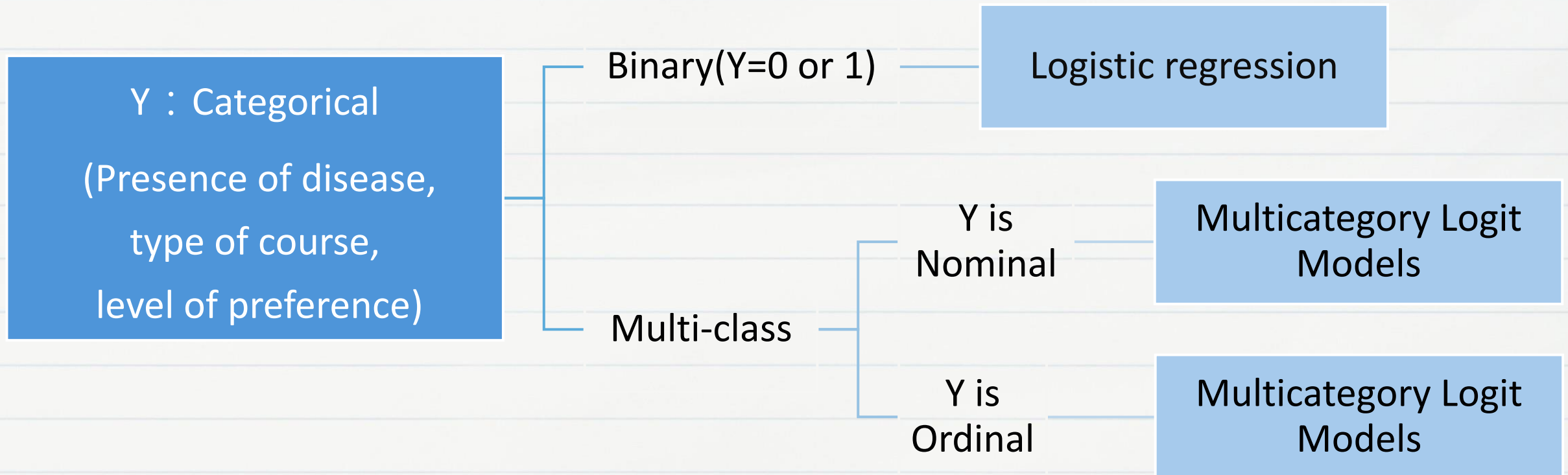
$$= \text{null deviance} - \text{residual deviance} \sim \chi^2_{df = \text{Number of variables used}}$$



# Categorical data

- Logistic Regression
- Logit models for Nominal Responses
- Logit models for Ordinal Responses

# GLM of Categorical data



# If the dataset we received today looks like this



- Data Overview : Basic information of 768 patients along with whether they have diabetes.
- Analysis Objective: To determine which variables may affect the likelihood of having diabetes.
- Variables: Y : **diabetes (whether the patient has diabetes)**

X : pregnant: number of pregnancies(懷孕次數)

glucose : plasma glucose concentration after 2 hours in an oral glucose tolerance test(血漿葡萄糖濃度)

blood pressure : diastolic blood pressure (mmHg)

triceps : skinfold thickness of the triceps (mm)

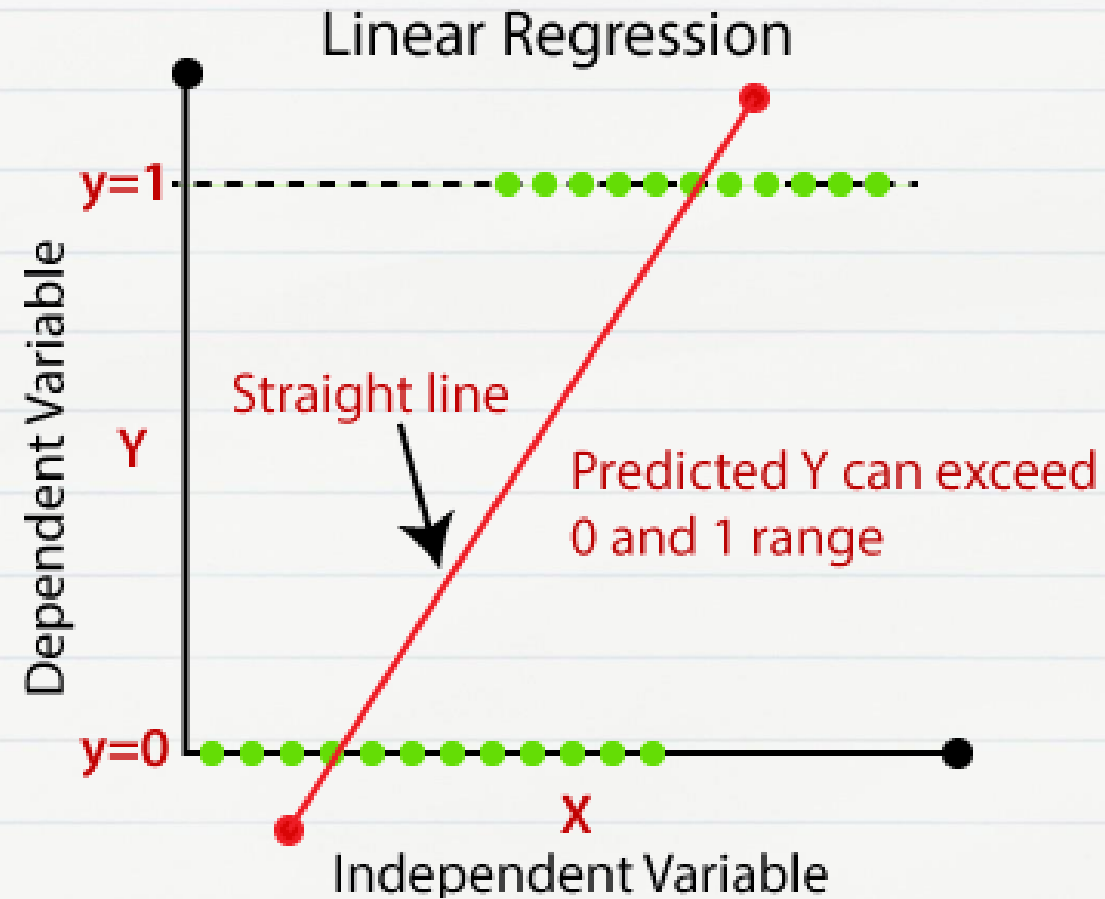
serum\_insulin : 2-hour serum insulin ( $\mu$ U/ml)

mass : BMI

pedigree : diabetes pedigree function (computed based on family diabetes history)

age : age

# If use Linear Regression



# Logistic Regression (羅吉斯迴歸)



- Timing of use :  $Y = 0$  or  $1$
- Logit link

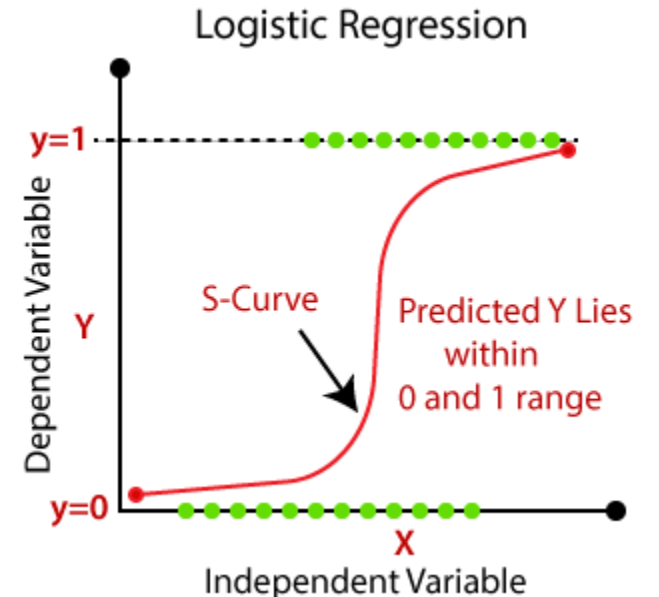
$$\pi(x) = P(Y = 1)$$

$$\text{logit}[\pi(x)] = \log\left(\frac{\pi(x)}{1-\pi(x)}\right) (\log(\text{odds})) = \alpha + \beta x$$

$$\rightarrow \pi(x) = \frac{\exp(\alpha + \beta x)}{1 + \exp(\alpha + \beta x)}$$

- $\frac{\text{odds at } x+1}{\text{odds at } x} = \frac{\exp(\alpha + \beta(x+1))}{\exp(\alpha + \beta x)} = e^\beta$

→ When  $x$  increases by one unit, the odds of the event happening become  $e^\beta$  times the original.



Sufficient sample size(n=768)  
→ Include all variables

Call:

```
glm(formula = diabetes ~ ., family = binomial(link = logit),  
     data = diabete)
```

Deviance Residuals:

| Min     | 1Q      | Median  | 3Q     | Max    |
|---------|---------|---------|--------|--------|
| -2.5566 | -0.7274 | -0.4159 | 0.7267 | 2.9297 |

Coefficients:

|                | Estimate   | Std. Error | z value  | Pr(> z ) |           |
|----------------|------------|------------|----------|----------|-----------|
| (Intercept)    | -8.4046964 | 0.7166359  | -11.728  | < 2e-16  | ***       |
| pregnant       | 0.1231823  | 0.0320776  | 3.840    | 0.000123 | ***       |
| glucose        | 0.0351637  | 0.0037087  | 9.481    | < 2e-16  | ***       |
| blood_pressure | -0.0132955 | 0.0052336  | -2.540   | 0.011072 | *         |
| triceps        | 0.0006190  | 0.0068994  | 0.090    | 0.928515 |           |
| serum_insulin  | -0.0011917 | 0.0009012  | -1.322   | 0.186065 |           |
| mass           | 0.0897010  | 0.0150876  | 5.945    | 2.76e-09 | ***       |
| pedigree       | 0.9451797  | 0.2991475  | 3.160    | 0.001580 | **        |
| age            | 0.0148690  | 0.0093348  | 1.593    | 0.111192 |           |
| ---            |            |            |          |          |           |
| Signif. codes: | 0 '***'    | 0.001 '**' | 0.01 '*' | 0.05 '.' | 0.1 ' ' 1 |

(Dispersion parameter for binomial family taken to be 1)

```
Null deviance: 993.48 on 767 degrees of freedom  
Residual deviance: 723.45 on 759 degrees of freedom  
AIC: 741.45
```

Number of Fisher Scoring iterations: 5

# Interpretation of coefficients

Coefficients:

|                | Estimate         | Std. Error | z value | Pr(> z ) |     |
|----------------|------------------|------------|---------|----------|-----|
| (Intercept)    | -8.4046964       | 0.7166359  | -11.728 | < 2e-16  | *** |
| pregnant       | <u>0.1231823</u> | 0.0320776  | 3.840   | 0.000123 | *** |
| glucose        | <u>0.0351637</u> | 0.0037087  | 9.481   | < 2e-16  | *** |
| blood_pressure | -0.0132955       | 0.0052336  | -2.540  | 0.011072 | *   |
| triceps        | 0.0006190        | 0.0068994  | 0.090   | 0.928515 |     |
| serum_insulin  | -0.0011917       | 0.0009012  | -1.322  | 0.186065 |     |
| mass           | 0.0897010        | 0.0150876  | 5.945   | 2.76e-09 | *** |
| pedigree       | 0.9451797        | 0.2991475  | 3.160   | 0.001580 | **  |
| age            | 0.0148690        | 0.0093348  | 1.593   | 0.111192 |     |

- pregnant ( $\beta_1 = 0.123, e^{\beta_1} = 1.131$ )

When holding all other variables constant, for each additional pregnancy, the odds of getting diabetes increase by a factor of 1.131.

- glucose ( $\beta_2 = 0.035, e^{\beta_2} = 1.036$ )

When holding all other variables constant, for each one-unit increase in glucose, the odds of getting diabetes increase by a factor of 1.036.

# Model Testing

|                           |        |                    |
|---------------------------|--------|--------------------|
| Null deviance: 993.48     | on 767 | degrees of freedom |
| Residual deviance: 723.45 | on 759 | degrees of freedom |

Likelihood Ratio Test

$$\begin{aligned} -2(L_0 - L_1) &= \text{null deviance} - \text{residual deviance} \\ &= 993.48 - 723.45 = 270.03 \sim \chi^2_{df=8} \end{aligned}$$

p-value < 0.0001

→ This model is statistically significant.

# If the dataset we received today looks like this

- Data Overview : The basic information of 200 students along with the chosen course types.
- Y : prog : program type (**general, academic, vocational**)
- X : ses : social economic status (low, middle, high) (社經地位)  
white : writing score (筆試分數)



# Logit Models for Nominal Responses (多項迴歸)

- Timing of use : Y has more than two categories, and the categories are nominal.
- Model :

Suppose Y has J categories

$$\pi_j = P(Y = j), j = 1, 2, \dots, J$$

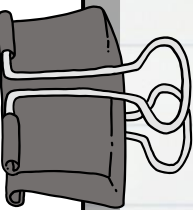
Baseline-category logits :

$$\log \left( \frac{\pi_j}{\pi_J} \right) = \alpha_j + \beta_j x, j = 1, \dots, J - 1$$

$\pi_J$  : baseline (The most interested or the sample with the largest size.)

$$\frac{\exp(\alpha_j + \beta_j(x+1))}{\exp(\alpha_j + \beta_j x)} = e^{\beta}$$

→ When x increases by 1-unit,  $\frac{P(Y=j)}{P(Y=J)}$  will be  $e^{\beta}$  times the original.



Call:

```
multinom(formula = prog2 ~ ses + write, data = data)
```

Coefficients:

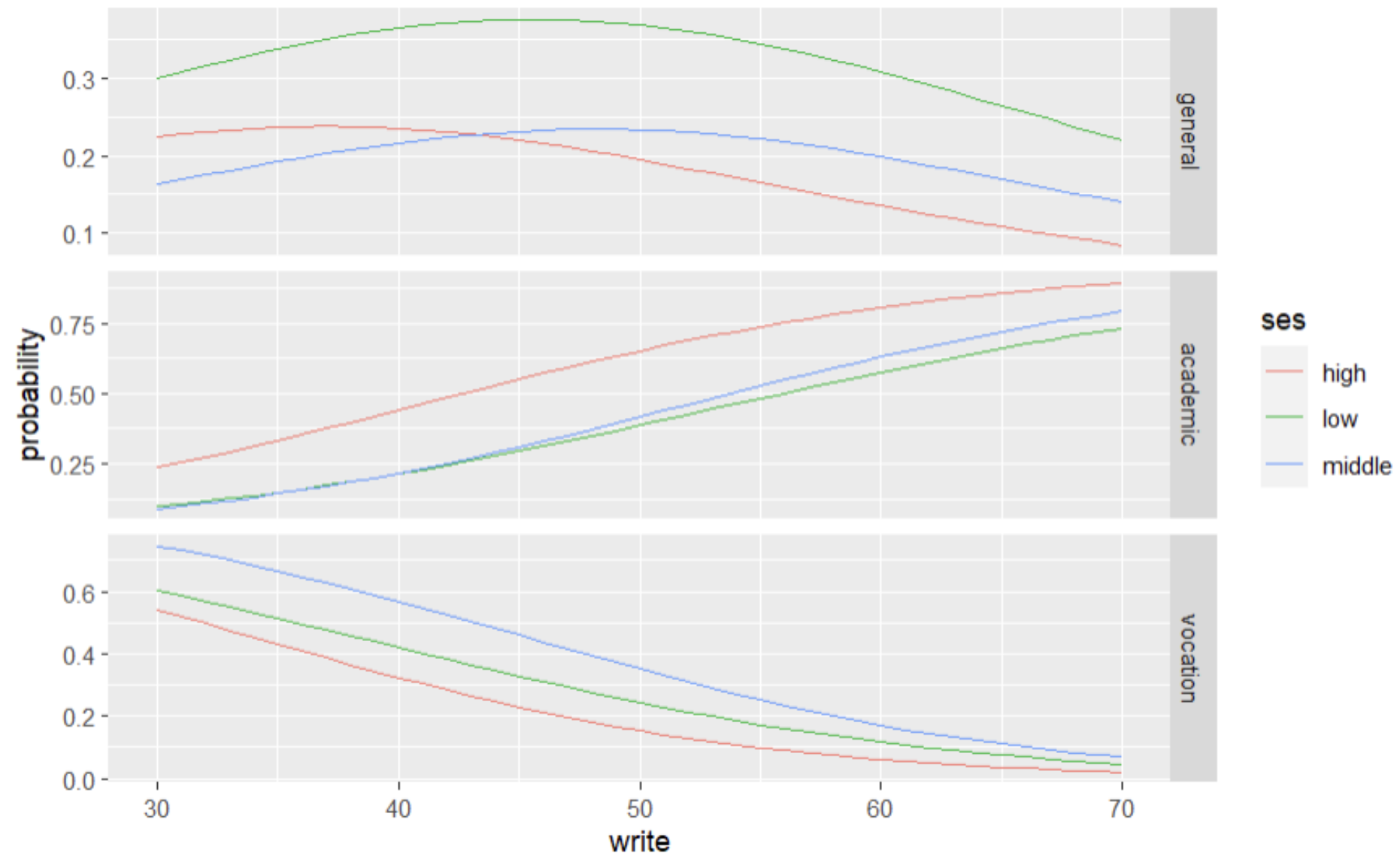
|          | (Intercept) | sesmiddle | seshigh   | write       |
|----------|-------------|-----------|-----------|-------------|
| academic | -2.851973   | 0.5332914 | 1.1628257 | 0.05792480  |
| vocation | 2.366097    | 0.8246384 | 0.1802176 | -0.05567514 |

$$\log\left(\frac{P(\text{academic})}{P(\text{general})}\right) = -2.85 + 0.53(\text{ses} = \text{middle}) + 1.16(\text{ses} = \text{high}) + 0.06\text{write}$$

$$\log\left(\frac{P(\text{vocation})}{P(\text{general})}\right) = 2.36 + 0.82(\text{ses} = \text{middle}) + 0.18(\text{ses} = \text{high}) - 0.06\text{write}$$

→ When ses=middle ,  $\frac{P(\text{academic})}{P(\text{general})}$  will be  $1.70(e^{0.53})$  times higher than when ses=low.

→ For every 1-unit increasing in write ,  $\frac{P(\text{academic})}{P(\text{general})}$  will be  $1.06(e^{0.06})$  times higher than the original.



# If the dataset we received today looks like this

- Y : apply : Willingness to apply to graduate school  
(unlikely, somewhat likely, very likely)
- X : pared : whether at least one parent has a graduate degree  
(父母是否有研究所學位)  
public : the undergraduate institution is public or private  
(1 : public, 0 : private) (就讀公立or私立學校)  
gpa : the student's grade point average (畢業分數)

# Cumulative Logit Models for Ordinal Responses (次序迴歸)

- Timing of use :  $Y$  has more than two categories, and the categories are ordinal.
- Model :

$$P(Y \leq j) = \pi_1 + \pi_2 + \cdots + \pi_j, j = 1, 2, \dots, J$$

$$\text{constrain : } \sum_{i=1}^n \pi_i = 1$$

$$\text{logit}[P(Y \leq j)](\log(odds)) = \alpha_j + \beta x, j = 1, 2, \dots, J - 1$$

(One  $x$  has only one  $\beta \rightarrow$  the effect on all categories of  $x$  is the same.)

$$\frac{\text{odds of } (Y \leq j) \text{ at } x+1}{\text{odds of } (Y \leq j) \text{ at } x} = \frac{\exp(\alpha_j + \beta(x+1))}{\exp(\alpha_j + \beta x)} = e^\beta$$

When  $x$  increases by 1-unit, the odds of  $Y \leq j$  become  $e^\beta$  times the original.

Call:

```
polr(formula = apply ~ pared + public + gpa, data = data, Hess = TRUE)
```

Coefficients:

|        | Value    | Std. Error | t value |
|--------|----------|------------|---------|
| pared  | 1.04769  | 0.2658     | 3.9418  |
| public | -0.05879 | 0.2979     | -0.1974 |
| gpa    | 0.61594  | 0.2606     | 2.3632  |

Intercepts:

|                             | Value  | Std. Error | t value |
|-----------------------------|--------|------------|---------|
| unlikely somewhat likely    | 2.2039 | 0.7795     | 2.8272  |
| somewhat likely very likely | 4.2994 | 0.8043     | 5.3453  |

$$\log(\text{odds of } Y \leq 1) = \log\left(\frac{P(Y \leq 1)}{P(Y > 1)}\right) = 2.20 - 1.05\text{pared} - (-0.06)\text{public} - 0.62\text{gpa}$$

$$\log(\text{odds of } Y \leq 2) = \log\left(\frac{P(Y \leq 2)}{P(Y > 2)}\right) = 4.30 - 1.05\text{pared} - (-0.06)\text{public} - 0.62\text{gpa}$$

→ Students whose parents has a graduate degree are  $2.85(e^{1.05})$  times more likely to apply for graduate school compared to students whose parents did not has a graduate degree.

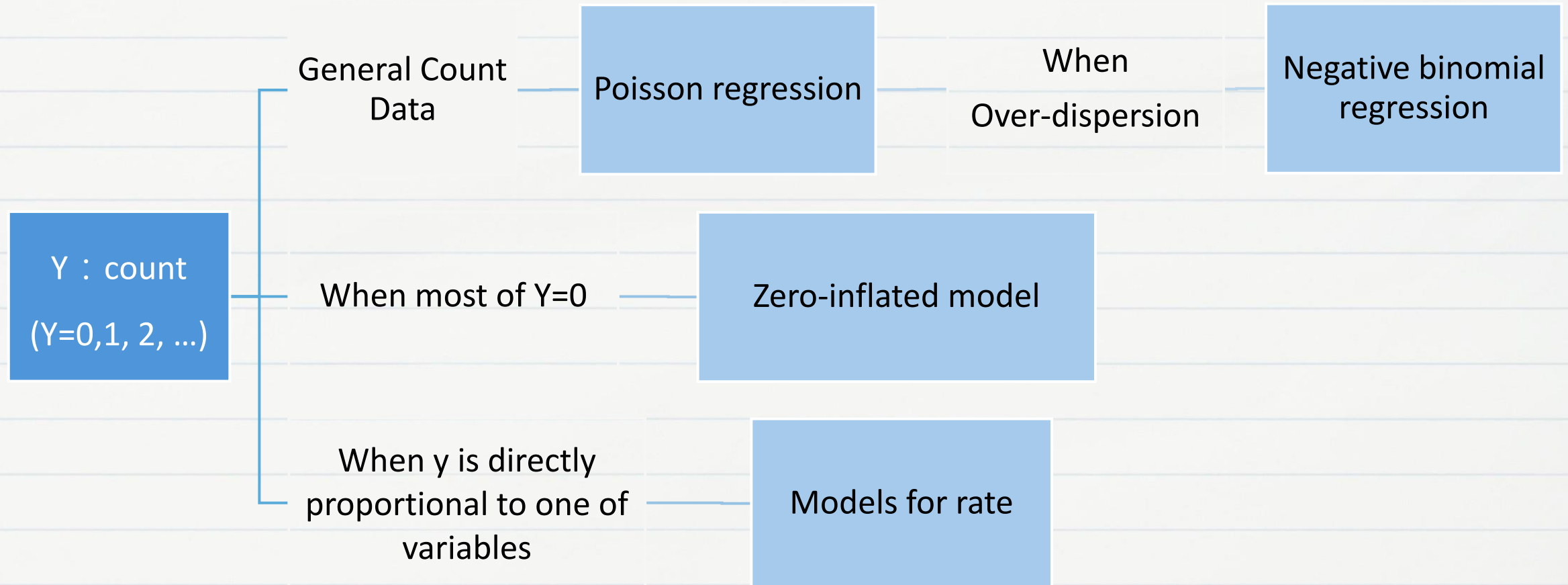
→ For every one-unit increase in GPA, students are  $1.85(e^{0.62})$  times more likely to apply for graduate school.



# Count Data

- Poisson Regression
- Negative Binomial Regression
- Zero-inflated model
- Models for rate

# GLM for Count data



# If the dataset we received today looks like this

- Data Overview : The number of awards received by 200 students at a certain high school along with other data
- Y : num\_awards : The number of awards received by each student in a year.  
(學生一年中獲得的獎項數量)
- X : prog : The type of course the student is enrolled in (General, Academic, Vocational). (選擇的課程類型)  
math : The student's score in the final math exam. (數學成績)

# Poisson Regression



- Poisson pmf

$$f(y; \mu) = \frac{e^{-\mu} \mu^y}{y!}, y = 0, 1, 2, \dots, \mu > 0$$

- Estimating the effect of  $x$  on  $y \rightarrow$  Estimating the effect of  $x$  on  $\mu$
- Identity link

$$\mu = \alpha + \beta x \quad (\text{Not guaranteed } \mu > 0)$$

- Log link

$$\log(\mu) = \alpha + \beta x \quad -\infty < \log(\mu) < \infty \quad (\text{Use MLE solve } \beta)$$

$$\rightarrow \mu = \exp(\alpha + \beta x) = \exp(\alpha) \times (\exp(\beta))^x$$

- When  $x$  increases by one unit,  $\mu$  becomes  $e^\beta$  times the original.

Call:

```
glm(formula = num_awards ~ math + prog, family = poisson(link = "log"),  
     data = awards)
```

Coefficients:

|                | Estimate | Std. Error | z value | Pr(> z ) |     |
|----------------|----------|------------|---------|----------|-----|
| (Intercept)    | -5.24712 | 0.65845    | -7.969  | 1.60e-15 | *** |
| math           | 0.07015  | 0.01060    | 6.619   | 3.63e-11 | *** |
| progAcademic   | 1.08386  | 0.35825    | 3.025   | 0.00248  | **  |
| progVocational | 0.36981  | 0.44107    | 0.838   | 0.40179  |     |

$Y \sim \text{Poisson}(\mu)$

$\log(\mu) = -5.25 + 0.07x_1 + 1.08x_2 + 0.37x_3$

$$x_1 = \text{math}, x_2 = \begin{cases} 1, & \text{if } \text{prog} = \text{Academic} \\ 0, & \text{Otherwise} \end{cases}, x_3 = \begin{cases} 1, & \text{if } \text{prog} = \text{Vocational} \\ 0, & \text{Otherwise} \end{cases}$$

→The higher the math score, the more awards received.

→Students who choose Academic program tend to receive more awards, while the other two programs show no significant difference.

# If the dataset we received today looks like this

- Y : nvics : In past 12 months, how many people have you know personally that were victims of homicide (凶殺案次數)
- X : race : Black, White (種族)

| Race \ Y | 0    | 1  | 2  | 3 | 4 | 5 | 6 |
|----------|------|----|----|---|---|---|---|
|          | 0    | 1  | 2  | 3 | 4 | 5 | 6 |
| Black    | 119  | 16 | 12 | 7 | 3 | 2 | 0 |
| White    | 1070 | 60 | 14 | 4 | 0 | 0 | 1 |

$\left\{ \begin{array}{l} \text{Black} : \bar{y} = 0.52, s^2 = 1.15 \\ \text{White} : \bar{y} = 0.09, s^2 = 0.16 \end{array} \right. \rightarrow \text{variance} > \text{mean} \rightarrow \text{Violates the Poisson assumption}$

# If use Poisson Regression

Call:

```
glm(formula = nvics ~ race, family = poisson(link = "log"), data = homicide,
     weights = Freq)
```

Deviance Residuals:

| Min     | 1Q    | Median | 3Q    | Max    |
|---------|-------|--------|-------|--------|
| -14.051 | 0.000 | 5.257  | 6.216 | 13.306 |

Coefficients:

|             | Estimate | Std. Error | z value | Pr(> z )     |
|-------------|----------|------------|---------|--------------|
| (Intercept) | -0.6501  | 0.1098     | -5.922  | 3.17e-09 *** |
| raceWhite   | -1.7331  | 0.1466     | -11.825 | < 2e-16 ***  |
| ---         |          |            |         |              |

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

Null deviance: 962.80 on 10 degrees of freedom  
Residual deviance: 844.71 on 9 degrees of freedom  
AIC: 1122

Excessive deviance

Number of Fisher Scoring iterations: 6

# Over-Dispersion (過度離散)



- Poisson model :  $\text{mean} = \text{variance}$
- But most of our data :  $\text{mean} < \text{variance} \rightarrow \text{overdispersion}$   
(if  $\text{mean} > \text{variance} \rightarrow \text{under-dispersion}$ )
- If still use Poisson model, residual will be large  
 $\rightarrow$  use negative binomial regression

# Negative Binomial Regression



- For negative binomial distribution  $\rightarrow$  variance  $>$  mean
- Set  $E(Y) = \mu, Var(Y) = \mu + D\mu^2$  ( $D$  : dispersion parameter)  
(Output in R display  $\theta = \frac{1}{D}$ )
- Link function  $\rightarrow \log(\mu) = \alpha + \beta x$

# Use Negative Binomial Regression

```
Call:
glm.nb(formula = nvics ~ race, data = homicide, weights = Freq,
       init.theta = 0.2023119205, link = log)
```

Deviance Residuals:

| Min     | 1Q    | Median | 3Q    | Max   |
|---------|-------|--------|-------|-------|
| -12.754 | 0.000 | 2.086  | 3.283 | 9.114 |

Coefficients:

|             | Estimate | Std. Error | z value | Pr(> z ) |     |
|-------------|----------|------------|---------|----------|-----|
| (Intercept) | -0.6501  | 0.2077     | -3.130  | 0.00175  | **  |
| raceWhite   | -1.7331  | 0.2385     | -7.268  | 3.66e-13 | *** |

The number of white people encountering murder cases is lower than that of black people.

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for Negative Binomial(0.2023) family taken to be 1)

Null deviance: 471.57 on 10 degrees of freedom  
Residual deviance: 412.60 on 9 degrees of freedom  
AIC: 1001.8

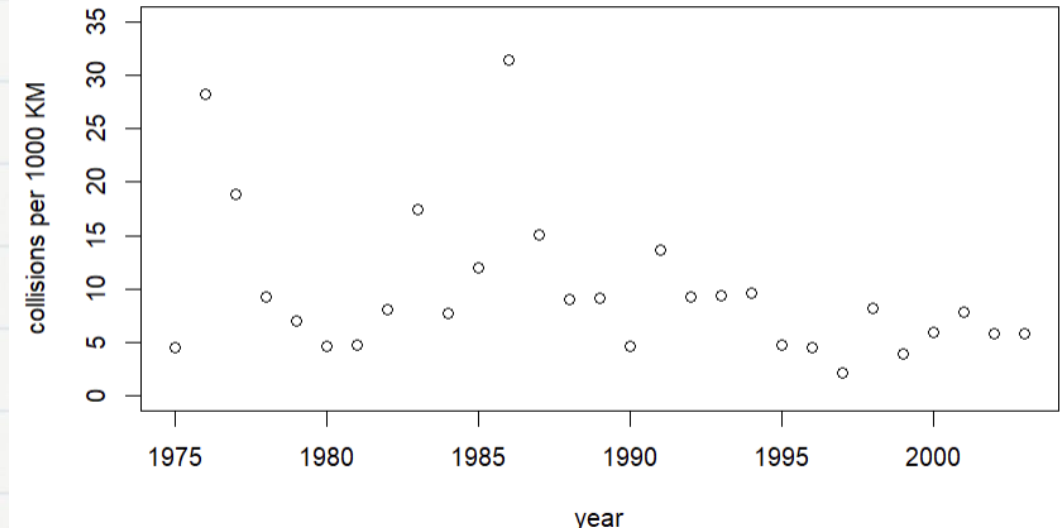
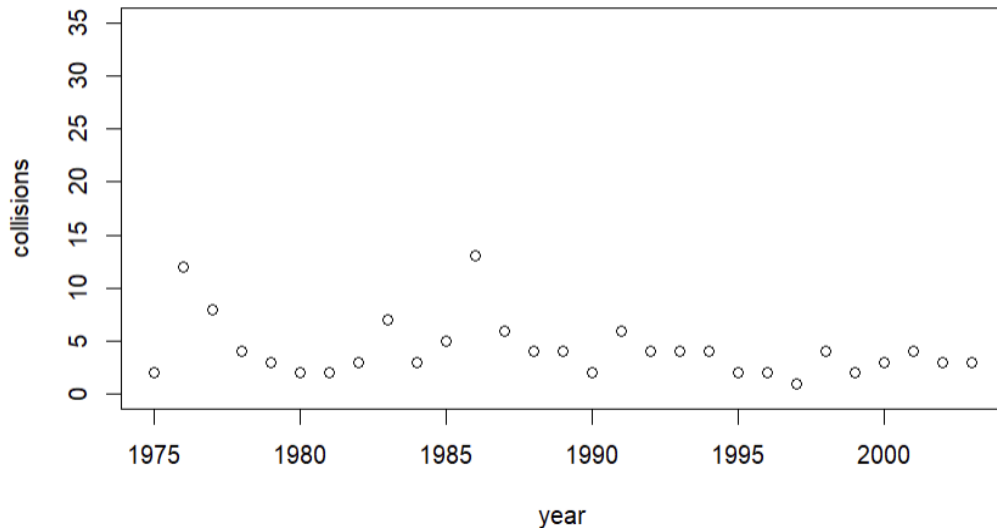
Number of Fisher Scoring iterations: 1

Theta: 0.2023  
Std. Err.: 0.0409

$$D = \frac{1}{\theta} \cong 4.94 > 0 \rightarrow \text{Var}(Y) = \mu + D\mu^2 = \mu + 4.94\mu^2$$

# If the dataset we received today looks like this

- We want to know whether the number of train accidents increases over time (as the year increases).
- Y : TrRd : The number of accidents involving trains and cars (發生事故次數)
- X : year (1975-2003) (年份)  
KM : The distance traveled by trains (KM) (火車行駛公里數)



# Models for Rate



- Timing of use : When  $y$  is directly proportional to one of variables( $t$ )
- Model : Assuming  $y$  is directly proportional to the variable  $t$

$$\text{sample rate } \frac{y}{t} \rightarrow E\left(\frac{Y}{t}\right) = \frac{\mu}{t}$$

$$\log\left(\frac{\mu}{t}\right) = \alpha + \beta x$$

$$\log(\mu) - \log(t) = \alpha + \beta x$$

↑ offset (adjusted term)

Call:

```
glm(formula = TrRd ~ I(Year - 1975), family = poisson(link = "log"),  
     data = train, offset = log(KM))
```

Deviance Residuals:

| Min     | 1Q      | Median  | 3Q     | Max    |
|---------|---------|---------|--------|--------|
| -2.0580 | -0.7825 | -0.0826 | 0.3775 | 3.3873 |

Coefficients:

|                | Estimate | Std. Error | z value | Pr(> z )    |
|----------------|----------|------------|---------|-------------|
| (Intercept)    | -4.21142 | 0.15892    | -26.50  | < 2e-16 *** |
| I(Year - 1975) | -0.03292 | 0.01076    | -3.06   | 0.00222 **  |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

$$\log\left(\frac{\mu}{t}\right) = -4.21 - 0.03x, x = year - 1975$$

→ As the year increases, the number of accidents decreases.



# Reference

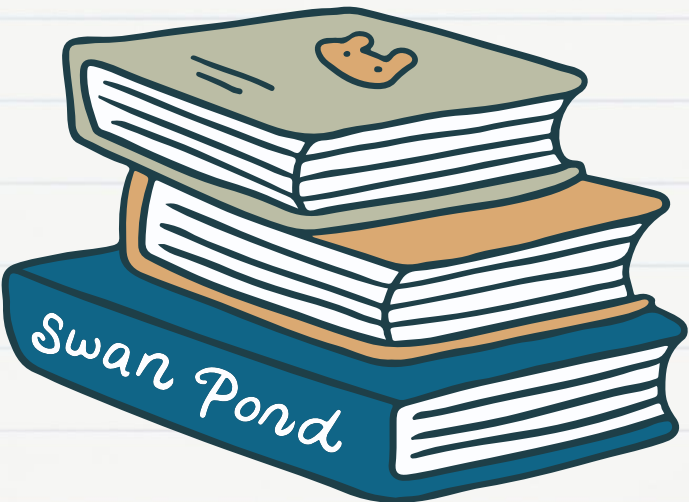
- ChatGPT : Translate my ppt into English
- 大學類別資料分析課程講義
- 羅吉斯迴歸 : <https://wenwu53.com/r-logistic-regression/>
- 多項迴歸 : <https://stats.oarc.ucla.edu/r/dae/multinomial-logistic-regression/>
- 次序迴歸 : <https://stats.oarc.ucla.edu/r/dae/ordinal-logistic-regression/>
- 計數型資料 : <https://stats.oarc.ucla.edu/other/dae/>



# Models for matched pairs

McNemar test & GEE

R26124039 徐淳姿



# Preliminary

- In GLM, we assume the data are independent
- But, there are other types of categorical data :
  - **Matched pairs** : each person answered two questions => **not** independent
    - To help the environment, would you be willing to (1) pay higher taxes (2) accept a cut in living standards ?
  - **Repeated measurement** : each person answered more than 2 questions => **not** independent
    - Whether you supported legalized abortion in the following three situations :
      - 1) If the family has very low income and cannot afford any more children
      - 2) When a woman is not married and does not want to marry a man
      - 3) When the woman wants it for any reason

How to analyze ?



# Matched pairs : McNemar test

- (Review) McNemar test : to examine if the proportions of categories in two related groups significantly differ from each other.

|                   | Cut on living standard (Q2) |                  |                  |
|-------------------|-----------------------------|------------------|------------------|
| Higher taxes (Q1) | YES                         | NO               | TOTAL            |
| YES               | 227 ( $n_{11}$ )            | 132 ( $n_{12}$ ) | 359 ( $n_{1+}$ ) |
| NO                | 107 ( $n_{21}$ )            | 678 ( $n_{22}$ ) | 785 ( $n_{2+}$ ) |
| TOTAL             | 334 ( $n_{+1}$ )            | 810 ( $n_{+2}$ ) | 1144             |

|                   | Cut on living standard (Q2) |                      |                      |
|-------------------|-----------------------------|----------------------|----------------------|
| Higher taxes (Q1) | YES                         | NO                   | TOTAL                |
| YES               | 0.198 ( $\pi_{11}$ )        | 0.115 ( $\pi_{12}$ ) | 0.313 ( $\pi_{1+}$ ) |
| NO                | 0.094 ( $\pi_{21}$ )        | 0.593 ( $\pi_{22}$ ) | 0.687 ( $\pi_{2+}$ ) |
| TOTAL             | 0.292 ( $\pi_{+1}$ )        | 0.708 ( $\pi_{+2}$ ) | 1                    |

$$P(Y_1 = 1) = \pi_{11} + \pi_{12}$$

$$P(Y_2 = 1) = \pi_{11} + \pi_{21}$$

$$H_0: P(Y_1 = 1) = P(Y_2 = 1)$$

$$\Rightarrow H_0: \pi_{12} = \pi_{21}$$

$$Z = \frac{n_{12} - \left(\frac{1}{2}\right)n^*}{\sqrt{n^* \left(\frac{1}{2}\right) \left(\frac{1}{2}\right)}} = \frac{n_{12} - n_{21}}{\sqrt{n_{12} + n_{21}}},$$

$$n^* = n_{12} + n_{21},$$

$$Z^2 \sim \chi_1^2$$



# Matched pairs : Confidence Interval

```
library(gee)
#install.packages("gee")
Opinions
summary(fit)
tab <- xtabs(~y1 + y2, data = Opinions)
tab
mcnemar.test(tab, correct=FALSE)
```

|    | y2  |     |
|----|-----|-----|
| y1 | 1   | 2   |
| 1  | 227 | 132 |
| 2  | 107 | 678 |

```
> mcnemar.test(tab, correct=FALSE)
```

McNemar's Chi-squared test

data: tab

McNemar's chi-squared = 2.6151, df = 1, p-value = 0.1059

## Confidence Interval

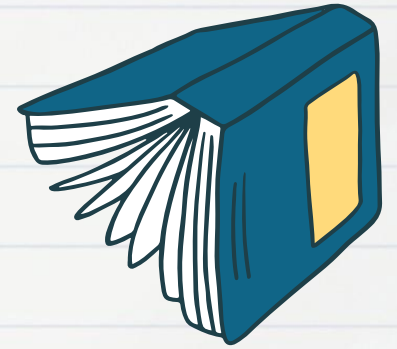
- $\hat{p} = P(Y_1 = 1) - P(Y_2 = 1) = 0.0219$
- $se(\hat{p}) = \frac{1}{n} \sqrt{(n_{12} + n_{21}) - (n_{12} - n_{21})^2 / n} = 0.0135$
- 95% Confidence interval :  $\hat{p} \pm 1.96 \times se(\hat{p})$

```
library(PropCIs)
#install.packages("PropCIs")
diffpropci.wald.mp(107, 132, 1144, 0.95)
```

95 percent confidence interval  
-0.004602847 0.048309140  
sample estimates:  
[1] 0.02185315 ( $\hat{p}$ )

→ No significant evidence to reject Null hypothesis (two proportion are the same)

# Matched pairs : marginal models

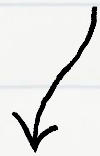


- No we consider using marginal models to examine the question

$$P(Y_1 = 1) = \alpha + \delta$$

$$P(Y_2 = 1) = \alpha$$

$$\delta = P(Y_1 = 1) - P(Y_2 = 1)$$



method : identity link

$$P(Y_1 = 1) = \alpha + \delta x_t$$

$$x_t = \begin{cases} 1, & t = 1 \\ 0, & t = 2 \end{cases}$$

$$\rightarrow H_0: \pi_{12} = \pi_{21}$$

$$\rightarrow H_0: \delta = 0$$

Alternative method : logit link

- $\text{logit}[P(Y_t = 1)] = \alpha + \beta x_t$
- $\beta$  : log odds ratio comparing two marginal distributions
- These **models** are called **marginal models**
- Focus on **marginal distributions of two responses**
- We use **generalized estimating equation** to fit the marginal models

# Why if we ignore correlation ?

- Take an example : there are two diet treatment, we want to test whether there is difference between two kinds of treatment ?

between group effect

with-in  
group  
effect

|                 | treatment A | treatment B |
|-----------------|-------------|-------------|
| Time 1 (before) | $y_1^A$     | $y_1^B$     |
| Time 2 (after)  | $y_2^A$     | $y_2^B$     |

Assuming there exists correlation

1)  $\text{Corr}(y_{i1}, y_{i2}) = \rho$

2)  $\text{Corr}(y_i, y_j) = 0$

3)  $\text{Corr}(y^A, y^B) = 0$

4)  $\text{Var}(y_{it}^A) = \text{Var}(y_{ju}^B) = \sigma^2$

$$y = \beta_0 + \beta_1 \times \text{treatment} + \beta_2 \times \text{Time}, \text{ where } \text{treatment} = \begin{cases} 1, \text{treatment A} \\ 0, \text{treatment B} \end{cases}, \text{time} = \begin{cases} 1, \text{time 1} \\ 0, \text{time 2} \end{cases}$$

• It can be shown that  $\hat{\beta}_1 = \frac{1}{2}(\bar{y}_1^A + \bar{y}_2^A - \bar{y}_1^B - \bar{y}_2^B)$ ,  $\hat{\beta}_2 = \frac{1}{2}(\bar{y}_1^A - \bar{y}_2^A + \bar{y}_1^B - \bar{y}_2^B)$

- And the variance is  $\text{Var}(\hat{\beta}_1) = \frac{\sigma^2(1+\rho)}{n}$ ,  $\text{Var}(\hat{\beta}_2) = \frac{\sigma^2(1-\rho)}{n}$
- If we ignore correlation:
  - under-estimated** the variance of **between group** effect (treatment)
  - over-estimated** the variance of **with-in** group effect (time)



# Marginal models : identity link

$$P(Y_1 = 1) = \alpha + \delta x_t$$

```
Opinions <- read.table("http://www.stat.ufl.edu/~aa/cat/data/Opinions.dat",
                      header=TRUE)
fit <- gee(y ~ question, id=person, family=binomial(link="identity"),
          data=Opinions) # id identifies variable on which observe y1, y2
summary(fit)
```

> summary(fit)

GEE: GENERALIZED LINEAR MODELS FOR DEPENDENT DATA  
gee S-function, version 4.13 modified 98/01/27 (1998)

Model:  
Link: Identity  
Variance to Mean Relation: Binomial  
Correlation Structure: Independent

Call:  
gee(formula = y ~ question, id = person, data = Opinions, family = binomial(link = "identity"))

Summary of Residuals:

| Min        | 1Q         | Median     | 3Q        | Max       |
|------------|------------|------------|-----------|-----------|
| -0.3138112 | -0.3138112 | -0.2919580 | 0.6861888 | 0.7080420 |

Coefficients:

|             | $\hat{\delta} (\hat{p})$ | $se(\hat{\delta})$ | Naïve     | Robust      |
|-------------|--------------------------|--------------------|-----------|-------------|
|             | Estimate                 | Naïve S.E.         | Naïve z   | Robust S.E. |
| (Intercept) | 0.29195804               | 0.01344828         | 21.709701 | 0.0134424   |
| question    | 0.02185315               | 0.01921587         | 1.137245  | 0.0134982   |

Estimated Scale Parameter: 1.000875  
Number of Iterations: 1

Working Correlation

|      | [,1] | [,2] |
|------|------|------|
| [1,] | 1    | 0    |
| [2,] | 0    | 1    |

|                   | Cut on living standard(Q2) |                    |                    |
|-------------------|----------------------------|--------------------|--------------------|
| Higher taxes (Q1) | YES                        | NO                 | TOTAL              |
| YES               | 227 ( $\pi_{11}$ )         | 132 ( $\pi_{12}$ ) | 359 ( $\pi_{1+}$ ) |
| NO                | 107 ( $\pi_{21}$ )         | 678 ( $\pi_{22}$ ) | 785 ( $\pi_{2+}$ ) |
| TOTAL             | 334 ( $\pi_{+1}$ )         | 810 ( $\pi_{+2}$ ) | 1144               |

$$P(Y_1 = 1) = \alpha + \delta = \frac{859}{1144} = 0.3138$$

$$P(Y_2 = 1) = \alpha = \frac{334}{1144} = 0.2919$$

$$\hat{\delta} = P(Y_1 = 1) - P(Y_2 = 1) = 0.0219$$

Naïve standard errors : assuming independent

Robust standard errors : assuming there's dependence

Most of matched pairs exhibit strong positive association

If odds ratio > 1 : robust < naïve → have association

# Marginal models : logit link

$$\text{logit}[P(Y_t = 1)] = \alpha + \beta x_t$$

```
fit2 <- gee(y ~ question, id=person, family=binomial(link=logit),
            data=Opinions)
summary(fit2)

> summary(fit2)

GEE:  GENERALIZED LINEAR MODELS FOR DEPENDENT DATA
gee S-function, version 4.13 modified 98/01/27 (1998)

Model:
Link:                               Logit
Variance to Mean Relation: Binomial
Correlation Structure:      Independent

Call:
gee(formula = y ~ question, id = person, data = Opinions, family = binomial(link = logit))

Summary of Residuals:
      Min       1Q   Median       3Q      Max
-0.3138112 -0.3138112 -0.2919580  0.6861888  0.7080420

Coefficients:       $\hat{\beta}$        $se(\hat{\beta})$ 
              Estimate Naive S.E.   Naive z Robust S.E.   Robust z
(Intercept) -0.8858933 0.06505597 -13.617401  0.06502753 -13.623357
question     0.1035319 0.09107816  1.136737  0.06397794  1.618244

Estimated Scale Parameter: 1.000875
Number of Iterations: 1

Working Correlation
      [,1] [,2]
[1,]    1    0
[2,]    0    1
```

|                   | Cut on living standard (Q2) |                  |                  |
|-------------------|-----------------------------|------------------|------------------|
| Higher taxes (Q1) | YES                         | NO               | TOTAL            |
| YES               | 227 ( $n_{11}$ )            | 132 ( $n_{12}$ ) | 359 ( $n_{1+}$ ) |
| NO                | 107 ( $n_{21}$ )            | 678 ( $n_{22}$ ) | 785 ( $n_{2+}$ ) |
| TOTAL             | 334 ( $n_{+1}$ )            | 810 ( $n_{+2}$ ) | 1144             |

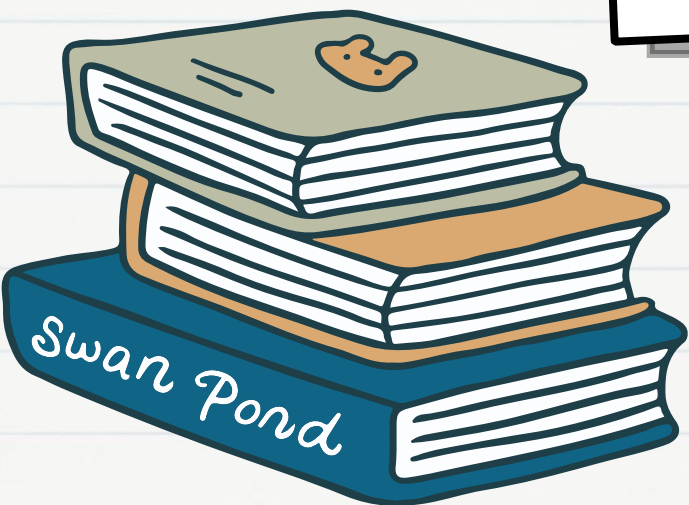
$$\hat{\beta} = \log \left[ \frac{359}{785} / \frac{334}{810} \right] = 0.104$$

$$\exp(\hat{\beta}) = 1.11$$

The population odds of willingness to pay higher taxes are estimated to be 11% higher than the population odds of willingness to accept cuts in living standards.



So what is GEE ?



# Generalized estimating equations (GEE)



- Proposed by Liang and Zeger (1986)
- In GLM, we assume the data are independent and from Normal
- GEE can be used to estimate the regression coefficient and variance of the data with repeated measurement that are not from Normal
- GEE is an estimating method rather than a model
- GEE doesn't need any assumed distribution on the data
- The marginal models are fitted by solving GEE



# GEE : example



- Whether you supported legalized abortion in the following three situations :
  - 1) If the family has very low income and cannot afford any more children
  - 2) When a woman is not married and does not want to marry a man
  - 3) When the woman wants it for any reason
- Response : record the responses with (situation 1, situation2, situation 3)

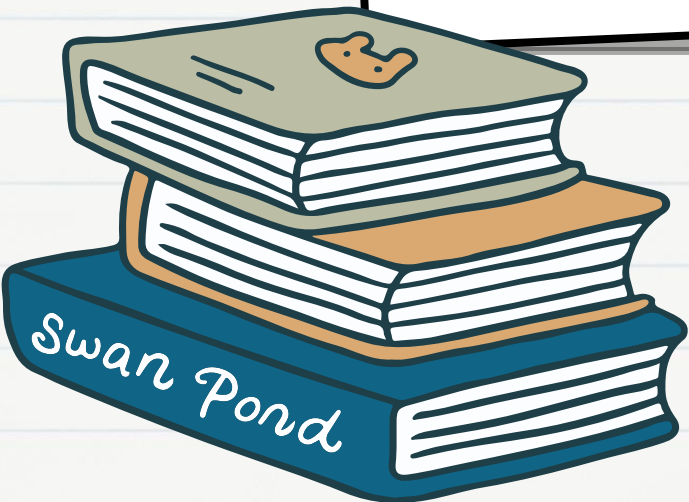
| response | (1,1,1) | (1,1,0) | (0,1,1) | (0,1,0) | (1,0,1) | (1,0,0) | (0,0,1) | (0,0,0) | total |
|----------|---------|---------|---------|---------|---------|---------|---------|---------|-------|
| Female   | 342     | 26      | 6       | 21      | 11      | 32      | 19      | 356     | 813   |
| Male     | 440     | 25      | 14      | 18      | 14      | 47      | 22      | 457     | 1037  |

- Data are dependent => use GEE !



# Estimating Equations?

Warning :  
the following 2 pages will  
include LOTS of math equations \*



# \*Estimating Equations

- When considering GLMs, we are majorly interested in the estimation and/or testing of the regression coefficients
- We may use MLE, write down the likelihood and take log
  - Then, take first-order derivative to the log-likelihood and set it to be 0
- Solve the equations and the root is the corresponding MLE estimators
- Is beginning with the likelihood the only way for estimation?
- **Estimating Equations** (first derivative of the log-likelihood is an example)



$$\sum_{i=1}^n \frac{\partial L(\boldsymbol{\beta}; y_i, x_i)}{\partial \boldsymbol{\beta}} = \sum_{i=1}^n \psi(\boldsymbol{\beta}; y_i, x_i) = 0$$

# \*Estimating Equations

\*(Bonus) Likelihood equations for GLM :

- $\frac{\partial L(\beta)}{\partial \beta_j} = \sum_{i=1}^n \frac{(y_i - \mu_i)x_i}{\text{var}(y_i)} \frac{\partial \mu_i}{\partial \eta_i} = 0, j = 1, 2, \dots, p$
- where  $\eta_i = \sum_{j=1}^p \beta_j x_{ij} = g(\mu_i)$ , for link function  $g$
- We rewrite the function with  $\mathbf{X}^T \mathbf{D} \mathbf{V}^{-1} (\mathbf{y} - \boldsymbol{\mu}) = \mathbf{0}$ ,  $\mathbf{D} = \frac{\partial \mu_i}{\partial \eta_i}$ ,  $\mathbf{V} = \text{var}(y_i)$
- Recall :  $\sum_{i=1}^n \frac{\partial L(\beta; y_i, x_i)}{\partial \beta} = \sum_{i=1}^n \psi(\beta; y_i, x_i) = 0$
- Then we can **rewrite** the estimating equation into
  - $\sum_{i=1}^n \psi(\beta; y_i, x_i) = \sum_{i=1}^n \frac{(y_i - \mu_i)x_i}{\text{var}(y_i)} \frac{\partial \mu_i}{\partial \eta_i} = \sum_{i=1}^n D_i^T V_i^{-1} (y_i - \mu_i) = 0$
  - where  $D_i = \frac{\partial \mu_i}{\partial \beta} = \frac{\partial \mu_i}{\partial \eta_i} \frac{\partial \eta_i}{\partial \beta_j} = \frac{\partial \mu_i}{\partial \eta_i} x_{ij}$  and  $V_i = \text{Var}(y_i)$   $\rightsquigarrow$

BUT we don't  
know the  
distribution of  $y_i$

# GEE structure

$$\eta_i = \sum_{j=1}^p \beta_j x_{ij} = g(\mu_i)$$

- Goal : Fit a model to repeated categorical responses, that is related of grouped response, by GEE methodology
- Variables :
  - Response variables ( $Y = (Y_{ij})$ ) : can be either binary or categorical
    - E.g.  $Y_{12}$  = the 2<sup>nd</sup> response of the first person
  - Predictors ( $X = (X_1, \dots, X_k)$ ) : can be discrete, continuous
    - E.g.  $X_1$  = the predictors of first person (gender)
- Model : Its form is like GLM, but we don't assume Y follow any distribution
 
$$g(\mu_i) = x_i^T \beta$$
- Link function : can be log link / logit link, etc.
- Solve to get  $\hat{\beta}$  :  $\sum_{i=1}^n \psi(\beta; y_i, x_i) = \sum_{i=1}^n \frac{(y_i - \mu_i)x_i}{\text{var}(y_i)} \frac{\partial \mu_i}{\partial \eta_i} = \sum_{i=1}^n D_i^T V_i^{-1} (y_i - \mu_i) = 0$

Since we don't know the distribution of  $y_i$

We will make a **specific correlation** guess among  $y_i$  → **Working Correlation matrix**

# GEE : working correlation

- We make a guess about covariance structure, called working correlation

|                                                                            |                                                                                                                                                                                  |                                                                                                                                                                                                                                     |                                                             |
|----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| independence                                                               | Identity matrix<br>$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$                                                              | Exchangeable correlation matrix<br>$\begin{pmatrix} 1 & \rho & \rho & \rho \\ \rho & 1 & \rho & \rho \\ \rho & \rho & 1 & \rho \\ \rho & \rho & \rho & 1 \end{pmatrix}$                                                             | $corr(Y_s, Y_t)$ is identical<br>for all pairs for all s, t |
|                                                                            | Autoregressive correlation matrix<br>$\begin{pmatrix} 1 & \rho & 0 & \rho^3 \\ \rho & 1 & \rho & \rho^2 \\ \rho^2 & \rho & 1 & \rho \\ \rho^3 & \rho^2 & \rho & 1 \end{pmatrix}$ | Unstructured correlation matrix<br>$\begin{pmatrix} 1 & \rho_{21} & \rho_{31} & \rho_{41} \\ \rho_{21} & 1 & \rho_{32} & \rho_{42} \\ \rho_{31} & \rho_{32} & 1 & \rho_{43} \\ \rho_{41} & \rho_{42} & \rho_{43} & 1 \end{pmatrix}$ |                                                             |
| With observations farther<br>apart in time being<br>more weakly correlated |                                                                                                                                                                                  |                                                                                                                                                                                                                                     | Permits each pair of<br>responses to be different           |

When we don't know the exact structure of correlation of responses, exchangeable correlation matrix would be a better guess. (Since it only need to estimate one parameter)

# GEE : example revisited



- Whether you supported legalized abortion in the following three situations :
  - 1) If the family has very low income and cannot afford any more children
  - 2) When a woman is not married and does not want to marry a man
  - 3) When the woman wants it for any reason

Response : record the responses with (situation 1, situation2, situation 3)

| response | (1,1,1) | (1,1,0) | (0,1,1) | (0,1,0) | (1,0,1) | (1,0,0) | (0,0,1) | (0,0,0) | total |
|----------|---------|---------|---------|---------|---------|---------|---------|---------|-------|
| Female   | 342     | 26      | 6       | 21      | 11      | 32      | 19      | 356     | 813   |
| Male     | 440     | 25      | 14      | 18      | 14      | 47      | 22      | 457     | 1037  |

- Data are dependent => use GEE !

# GEE : example revisited

- There are 813 females, 1037 males in this study (each of them response 3 times)
- Let female = 0, male = 1 (gender column)
- There are 3 situations (situations column)
- Let no = 0, yes = 1 (response column)
- Each person have 3 responses
- The marginal model is
$$\text{logit}[P(Y_t = 1)] = \alpha + \beta_t + \gamma x$$
- t = different situations
- x = gender
- We set the baseline is situation 3

| > Abortion |        |        |           |          |
|------------|--------|--------|-----------|----------|
|            | person | gender | situation | response |
| 1          | 1      | 1      | 1         | 1        |
| 2          | 1      | 1      | 2         | 1        |
| 3          | 1      | 1      | 3         | 1        |
| 4          | 2      | 1      | 1         | 1        |
| 5          | 2      | 1      | 2         | 1        |
| 6          | 2      | 1      | 3         | 1        |
| 7          | 3      | 1      | 1         | 1        |
| 8          | 3      | 1      | 2         | 1        |
| 9          | 3      | 1      | 3         | 1        |
| 10         | 4      | 1      | 1         | 1        |

情境一：低薪家庭，已經無法再負擔小孩的支出  
情境二：非自願受孕  
情境三：受孕者希望墮胎（無特定理由）

# GEE : example revisited

- We try to fit the model using GLM (**assuming** all the responses are **independent**)
- There's  $(813+1037)*3 = 5550$  responses, which is equal to there are 5550 people answer the survey

```
Abortion <- read.table("http://www.stat.ufl.edu/~aa/cat/data/Abortion.dat",  
                      header=TRUE)
```

```
Abortion |  
sit <- factor(Abortion$situation, levels=c(3,1,2))  
fit.glm <- glm(response ~ sit + gender, family=binomial, data=Abortion)  
summary(fit.glm) # ML estimates for 5550 independent observations
```

Baseline = 3

```
> summary(fit.glm) # ML estimates for 5550 independent observations
```

Call:

```
glm(formula = response ~ sit + gender, family = binomial, data = Abortion)
```

Coefficients:

|             | Estimate  | Std. Error | z value | Pr(> z ) |
|-------------|-----------|------------|---------|----------|
| (Intercept) | -0.125408 | 0.055601   | -2.255  | 0.0241 * |
| sit1        | 0.149347  | 0.065825   | 2.269   | 0.0233 * |
| sit2        | 0.052018  | 0.065843   | 0.790   | 0.4295   |
| gender      | 0.003582  | 0.054138   | 0.066   | 0.9472   |

---  
Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 7689.5 on 5549 degrees of freedom  
Residual deviance: 7684.2 on 5546 degrees of freedom  
AIC: 7692.2

Number of Fisher Scoring iterations: 3

# GEE : example

- We set the working correlation matrix is identity ( assuming all the responses are independent)
- The naïve s.e. here is equal to using glm, where robust s.e. consider **the id (same person) effect**
- ➔ robust s.e. consider group effect
- Situation : Naïve s.e. > Robust s.e.
- Gender : Naïve s.e. < Robust s.e.

Why there's difference ?

```
library(gee)
fit.gee <- gee(response ~ sit + gender, id=person, family=binomial,
               constr="independence" data=Abortion) # cluster on "id" variable
summary(fit.gee)
```

GEE: GENERALIZED LINEAR MODELS FOR DEPENDENT DATA  
gee S-function, version 4.13 modified 98/01/27 (1998)

Model:  
Link: Logit  
Variance to Mean Relation: Binomial  
Correlation Structure: Independent

Call:  
gee(formula = response ~ sit + gender, id = person, data = Abortion,  
family = binomial, constr = "independence")

Summary of Residuals:

|  | Min        | 1Q         | Median     | 3Q        | Max       |
|--|------------|------------|------------|-----------|-----------|
|  | -0.5068800 | -0.4825552 | -0.4686891 | 0.5174448 | 0.5313109 |

Coefficients:

|             | Estimate     | Naïve S.E. | Naïve z     | Robust S.E. | Robust z    |
|-------------|--------------|------------|-------------|-------------|-------------|
| (Intercept) | -0.125407576 | 0.05562131 | -2.25466795 | 0.06758236  | -1.85562596 |
| sit1        | 0.149347113  | 0.06584875 | 2.26803253  | 0.02973865  | 5.02198759  |
| sit2        | 0.052017989  | 0.06586692 | 0.78974374  | 0.02704704  | 1.92324166  |
| gender      | 0.003582051  | 0.05415761 | 0.06614123  | 0.08784012  | 0.04077921  |

Estimated Scale Parameter: 1.000721  
Number of Iterations: 1

Working Correlation

|      | [,1] | [,2] | [,3] |
|------|------|------|------|
| [1,] | 1    | 0    | 0    |
| [2,] | 0    | 1    | 0    |
| [3,] | 0    | 0    | 1    |

# GEE : With-in group vs Between group

- group : We define a person as a group
- With-in group : three responses in different situations
- Between group : gender
- s.e. for **with-in** group effects tend to be **smaller** than assuming the observations are independent (recall p. 6)
- s.e. for **between** group effects tend to be **larger** than assuming the observations are independent



Coefficients:

|             | Estimate     | Naive S.E. | Robust S.E. |
|-------------|--------------|------------|-------------|
| (Intercept) | -0.125407576 | 0.05562131 | 0.06758236  |
| sit1        | 0.149347113  | 0.06584875 | 0.02973865  |
| sit2        | 0.052017989  | 0.06586692 | 0.02704704  |
| gender      | 0.003582051  | 0.05415761 | 0.08784012  |

# GEE : example

- We set the working correlation matrix is exchangeable ( assuming the correlation pf three responses are the same)
- Naïve s.e. is similar to robust s.e., which means this correlation structure capture sample correlation well.
- Explanation :
  - $\hat{\beta}_1 = 0.149, e^{\hat{\beta}_1} = 1.16$   
 ➔ The estimated odds of supporting legalized abortion in situation 1 equal 1.16 times the estimated odds in situation 3
  - $\hat{\gamma} = 0.003, e^{\hat{\gamma}} = 1.003$   
 ➔ for each situation probability of supporting legalized abortion is similar for females and males

```
fit.gee2 <- gee(response ~ sit + gender, id=person, family=binomial,
               corstr="exchangeable", data=Abortion)
summary(fit.gee2)
```

GEE: GENERALIZED LINEAR MODELS FOR DEPENDENT DATA  
 gee S-function, version 4.13 modified 98/01/27 (1998)

Model:

Link: Logit  
 Variance to Mean Relation: Binomial  
 Correlation Structure: Exchangeable

Call:

```
gee(formula = response ~ sit + gender, id = person, data = Abortion,
    family = binomial, corstr = "exchangeable")
```

Summary of Residuals:

| Min        | 1Q         | Median     | 3Q        | Max       |
|------------|------------|------------|-----------|-----------|
| -0.5068644 | -0.4825396 | -0.4687095 | 0.5174604 | 0.5312905 |

Coefficients:

|             | Estimate     | Naive S.E. | Naive z     | Robust S.E. | Robust z    |
|-------------|--------------|------------|-------------|-------------|-------------|
| (Intercept) | -0.125325730 | 0.06782579 | -1.84775925 | 0.06758212  | -1.85442135 |
| sit1        | 0.149347107  | 0.02814374 | 5.30658404  | 0.02973865  | 5.02198729  |
| sit2        | 0.052017986  | 0.02815145 | 1.84779075  | 0.02704703  | 1.92324179  |
| gender      | 0.003437873  | 0.08790630 | 0.03910838  | 0.08784072  | 0.03913758  |

Estimated Scale Parameter: 1.000721

Number of Iterations: 2

Working Correlation

|      | [,1]      | [,2]      | [,3]      |
|------|-----------|-----------|-----------|
| [1,] | 1.0000000 | 0.8173308 | 0.8173308 |
| [2,] | 0.8173308 | 1.0000000 | 0.8173308 |
| [3,] | 0.8173308 | 0.8173308 | 1.0000000 |

$\hat{\rho} = 0.8173$

# GEE : Conclusion, Benefit & Drawbacks

- GEE
  - Deal with repeated measurement data
  - Data not independent, not from Normal
  - Response variable (Y) is categorical / binary
  - Should consider the correlation between response  
→ working correlation matrix

- Benefit & Drawbacks

| Benefit                                                                                                                                                               | Drawbacks                                                                                                                          |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>• No assumption of distribution</li><li>• Calculation is easier than MLE</li><li>• Can deal with repeated measurement</li></ul> | <ul style="list-style-type: none"><li>• No assumption of distribution =&gt; cannot do the testing that need distribution</li></ul> |



# Reference

- Agresti, A. (2012). *Categorical data analysis* (Vol. 792). John Wiley & Sons.
- Agresti, A. (2015). *Foundations of linear and generalized linear models*. John Wiley & Sons.
- [https://ods.tmu.edu.tw/portal\\_c3\\_cnt.php?owner\\_num=c3\\_71137&button\\_num=c3&folder\\_id=4616](https://ods.tmu.edu.tw/portal_c3_cnt.php?owner_num=c3_71137&button_num=c3&folder_id=4616)
- Yu Shyr (2022) ° Textbook of advanced biostatistics °
- [https://www.youtube.com/watch?v=ROdkmShnwoA&t=553s&ab\\_channel=MeerkatStatistics](https://www.youtube.com/watch?v=ROdkmShnwoA&t=553s&ab_channel=MeerkatStatistics)



Thank you  
for your listening!

